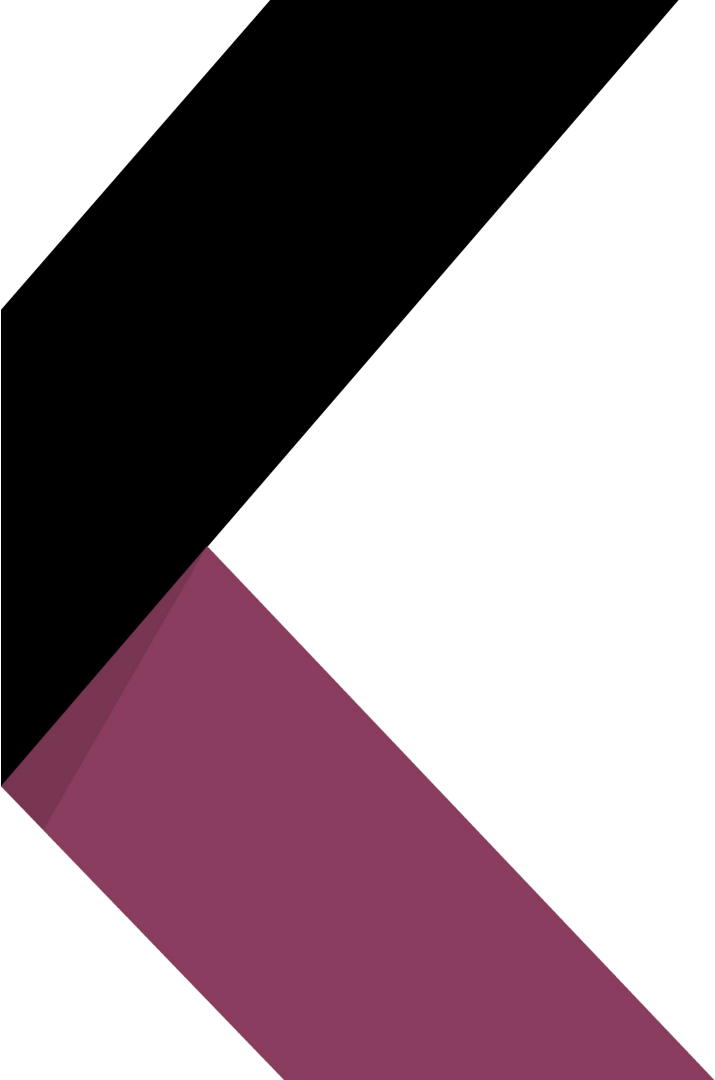


Electric Machines

A decorative graphic on the right side of the slide. It consists of a large black triangle pointing downwards, with a smaller maroon triangle nested inside its lower portion. The maroon triangle is also pointing downwards and is slightly offset to the right relative to the black triangle.



Introduction

Electric Machines - Overview

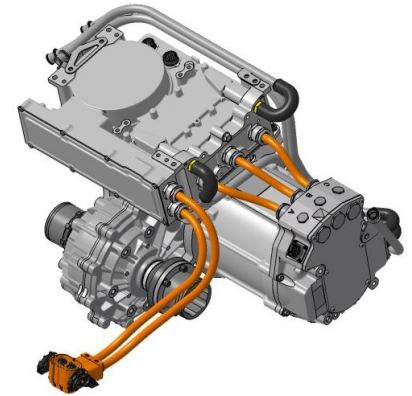
- Electric machines play a crucial role in all industry fields and power generation, as well as in automotive traction. In this modern reality and under the global need for CO₂ emission reduction, electric motors and generators are continuously evolving, aiming for higher efficiencies simultaneously with more reliable operation.
- Motor drives are the core technology for EVs that convert the on-board electrical energy to the desired mechanical motion. Meanwhile, electric machines are the key element of motor drive technology. The requirements of electric machines for EVs are much more demanding than that for industrial applications.



Electrical machines – Automotive demanding requirements

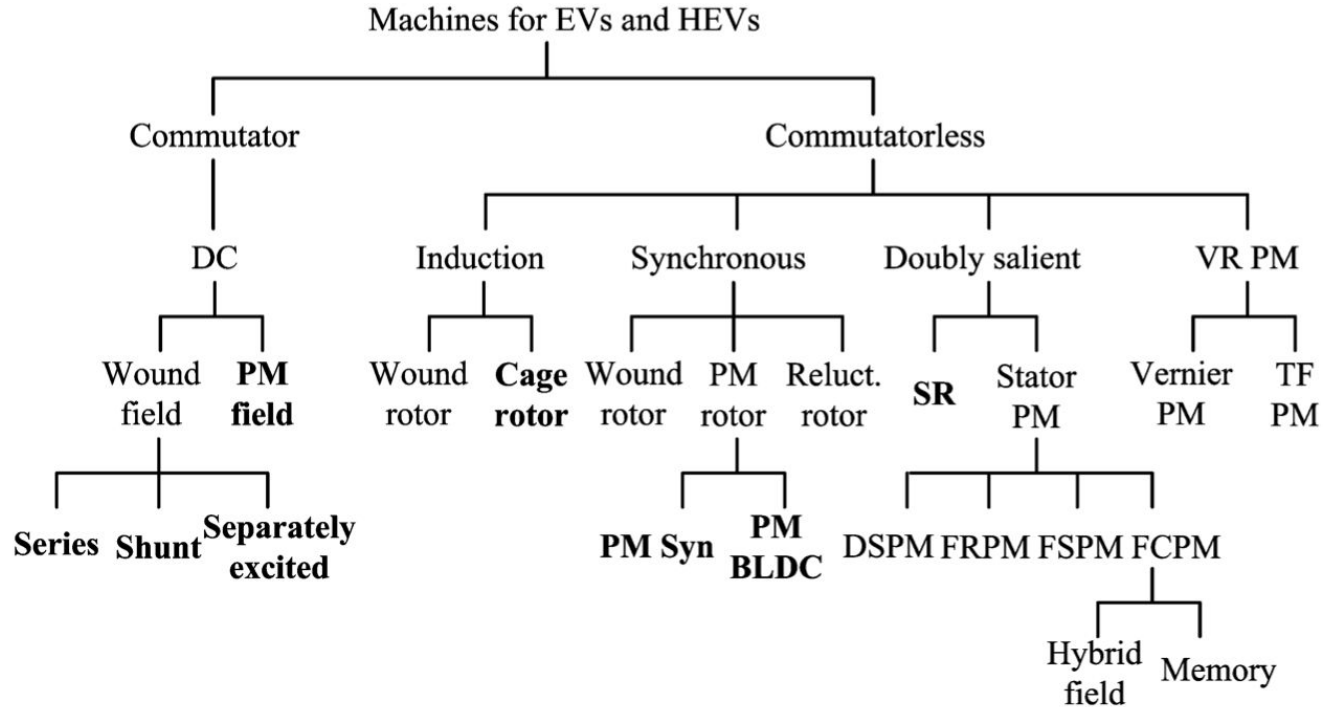
The general requirements of electrical machines for automotive are much more stringent than those for industrial applications:

- High torque density and high power density;
- Wide speed range, covering low-speed creeping and high-speed cruising;
- High efficiency over wide torque and speed ranges;
- Wide constant power operating capability;
- Low acoustic noise;
- Reasonable cost.





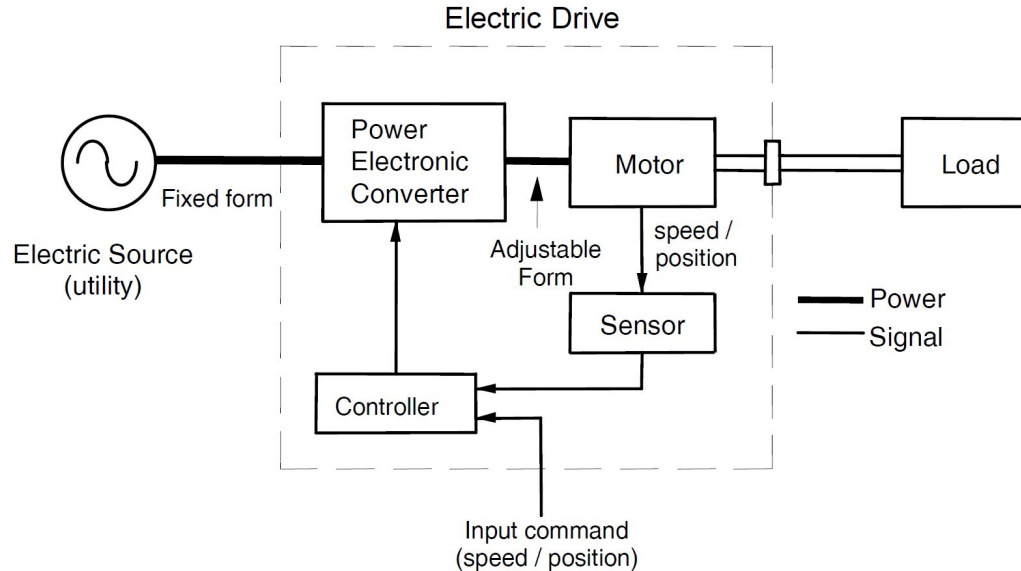
Electrical machines - Classification





Electric Drive

An electric drive can be defined as an electromechanical device for converting **electrical energy into mechanical energy** to impart motion to different machines and mechanisms for various kinds of process control.

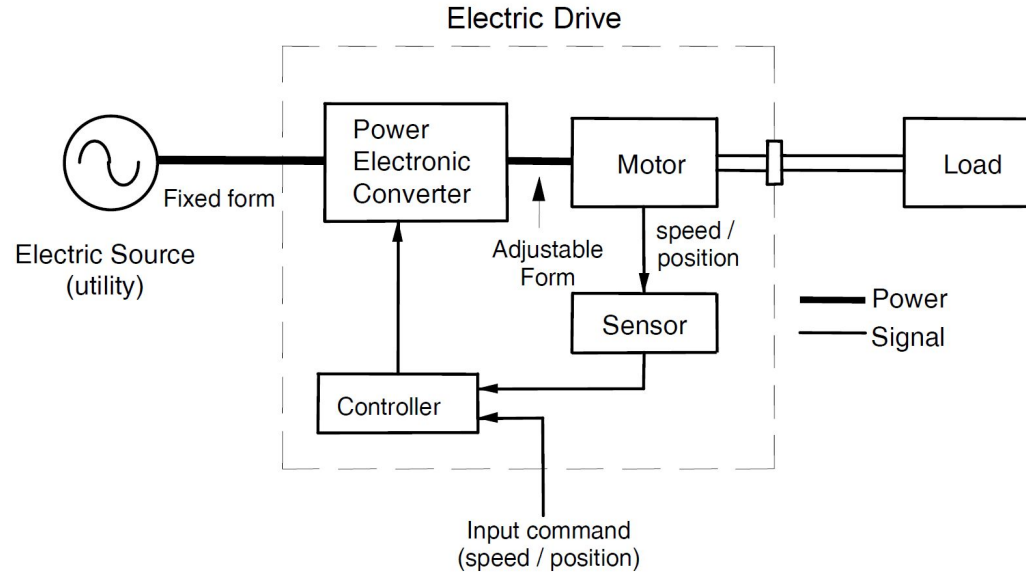




Components of an electric drive

The basic architecture of a motion control system includes:

1. **Power source:** provides the energy to the drive system
2. **Power converter:** interfaces the motor with the power source and provides the motor with adjustable voltage, current and/or frequency.
3. **Controller:** by comparing the input command for speed and/or position with the actual values measured through sensors, provides appropriate control signals to the power converter.



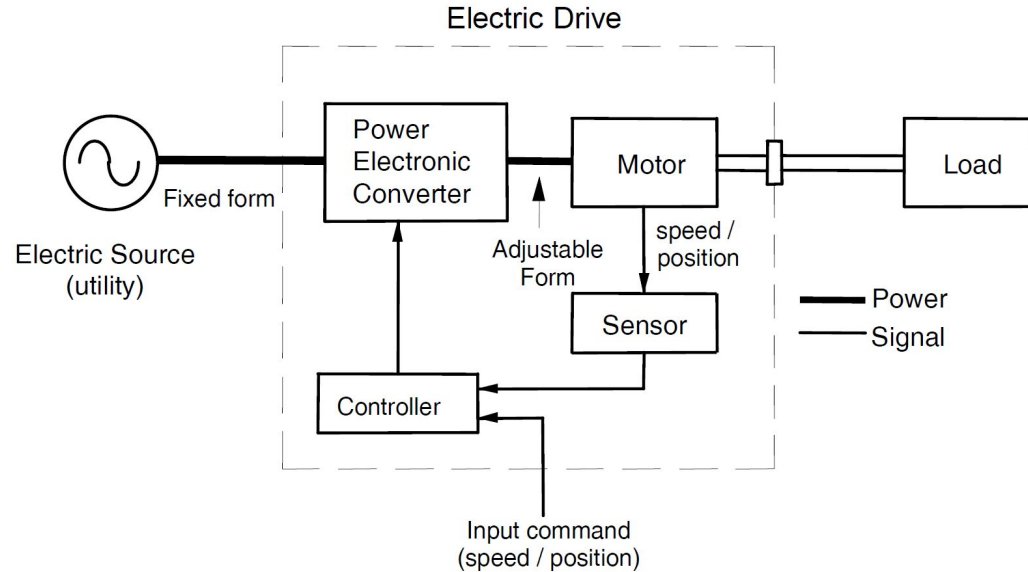


Components of an electric drive

4. **Electric motor**: selected according to the power level, environmental factors and performance required by the load.

5. One or more **feedback sensors** to return the position and/or velocity of the actuator to the motion controller in order to close the position and/or velocity control loops.

6. **Mechanical components** to transform the motion of the actuator into the desired load motion, including: gears, shafting, ball screw, belts, linkages, and linear and rotational bearings.





Basic principles



Basic principles and quantities

Magnetic flux, Φ , can be interpreted as a number of magnetic lines of force that cross certain given area - the greater the number of lines, the greater the flux and stronger the field.

Magnetic force

A current carrying conductor in presence of a magnetic field is exposed to a force equal to:

$$\mathbf{F} = i(\mathbf{l} \times \mathbf{B})$$

Induced electromotive force

Consider a rectangular loop rotating in a uniform magnetic field $\mathbf{B}$.

Time-varying flux causes an induced **electromotive force (e.m.f.)**, $\mathbf{e}$, at the terminals of a closed loop (or, a single conductor):

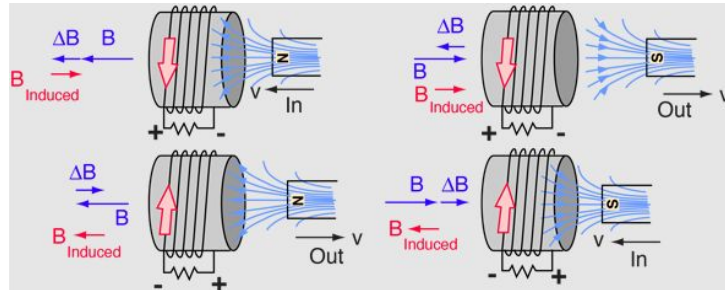
$$e = - \frac{d\phi}{dt}$$



Electromotive force

In a motor, the induced emf always acts to reduce the current in the loop. That's why it is called “**back emf.**”

In fact, Faraday-Neumann-Lenz law establish that *the induced current in a loop is in the direction that creates a magnetic field that **opposes the change** in magnetic flux through the area enclosed by the loop.* The induced current tends to keep the original magnetic flux through the circuit from changing.





Basic principles and quantities

Torque

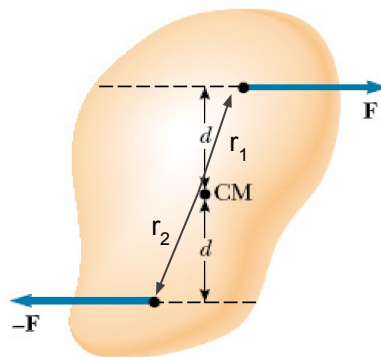
A **couple** is a pair of forces, $\mathbf{F}$ and $-\mathbf{F}$, equal in magnitude, opposite in direction, and separated by a distance, d . It can cause a rotation

The moment of a force $\mathbf{F}$ about a point O is the vector product of $\mathbf{F}$ and the position vector, $\mathbf{r}$, from O to the point of action of the force vector $\mathbf{M} = \mathbf{r} \times \mathbf{F}$

The moment of a couple:

$$\mathbf{M} = \mathbf{r}_1 \times \mathbf{F} + \mathbf{r}_2 \times (-\mathbf{F}) = (\mathbf{r}_1 - \mathbf{r}_2) \times \mathbf{F} = \boldsymbol{\tau}$$

is called **torque**, $\boldsymbol{\tau}$.



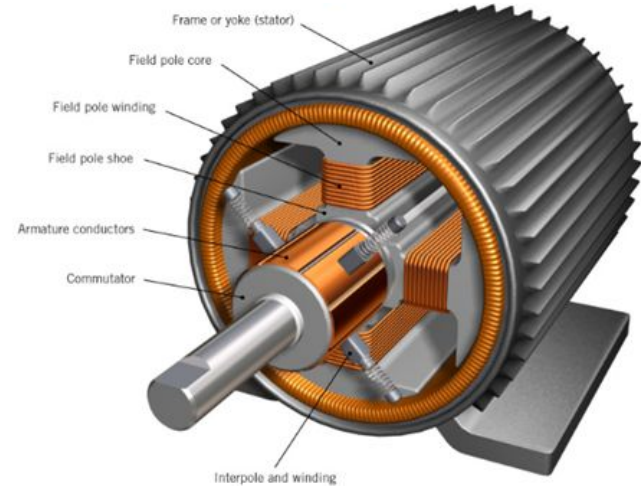


Electric Motors

An electric motor converts electrical energy to mechanical energy. It consists in a cylindrical structure with a fixed, static part, called **stator**, and a moving, rotating one, the **rotor**.

Both the structures can be populated with either **windings** or **permanent magnets**, achieving different electrical and magnetic performances.

The energy conversion happens in the **interaction between the magnetic fields** of the two structures, which produces a net output torque.





DC Machine

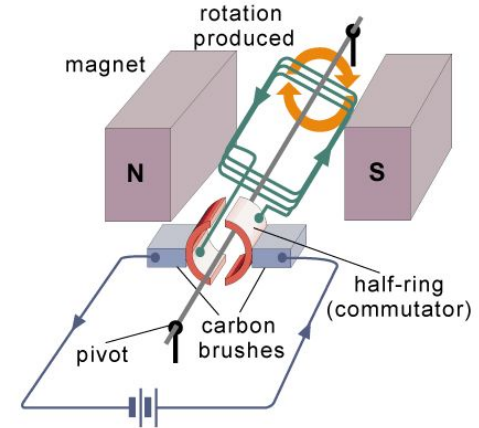


DC Machine

The **stator**, external, equipped with salient poles, works as an inductor and can be realized:

- with **excitation windings** arranged around the pole bodies; such windings are crossed by direct current and are connected to each other in controversial series to generate magnetic polarity of alternating sign from one pole to the next;
- with **permanent magnets**.

The internal **rotor**, consisting of a cylinder of magnetic material, works as armature; on the periphery of the rotor are uniformly distributed slots where the active conductors are housed. The set of active conductors (connected in series with each other) constitutes a closed circuit.





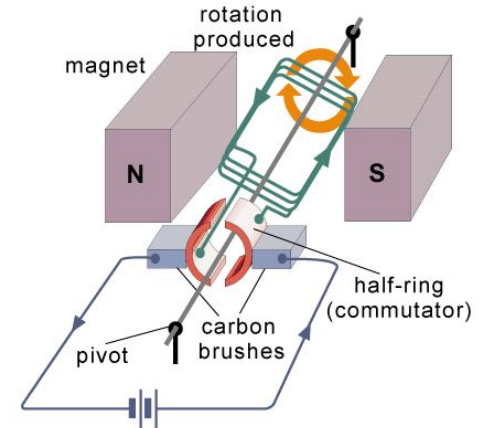
DC Machine

The part in the air between stator and rotor is called **air gap**, in this area the field magnetic has the greatest intensity.

The **commutator** is mounted on the same shaft as the rotor. This is for the purpose of supply the windings with the power supply voltage and ensure that the torque generated by the current that passes through is constant or at least unidirectional.

The **brushes** crawl on the collector and electrically connect it to the external terminals of the machine.

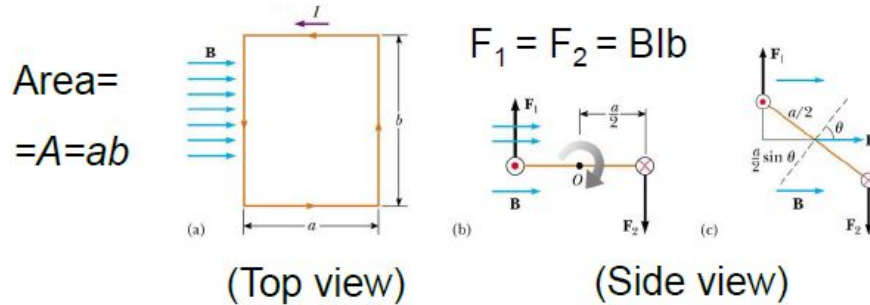
The presence of the system commutator-brushes causes various issues, mainly the continuous maintenance required and the undesired sparking.





DC Machine – Working principle

Consider a rectangular conducting loop carrying current I in the presence of a magnetic field:



There is a magnetic force on sides of length b (in opposite directions).

If loop can pivot about its center, these forces provide a (clockwise) **torque** about the pivot point.

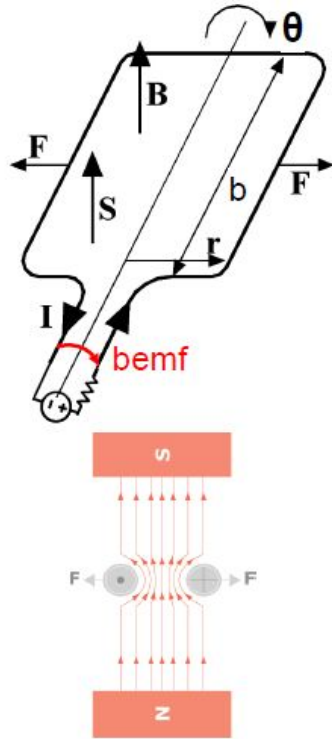
Maximum torque produced is $\tau_{max} = B I b \left(\frac{a}{2} + \frac{a}{2} \right) = B I A$

If field makes an angle θ with a line perpendicular to the plane of the loop then $\tau = B I A \sin \theta$

For any loop shape having N turns: $\tau = N B I A \sin \theta$



Conducting loop in a magnetic field



$$\theta = \omega t = 0$$

$$\Phi = \vec{S} \cdot \vec{B} = 2rbB \cos(\theta) = 2rbB = \Phi_{\text{MAX}}$$

$$\vec{F} = b\vec{I} \times \vec{B} \rightarrow F = bIB$$

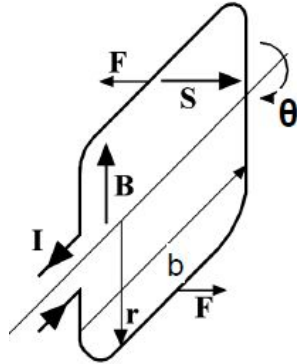
$$\vec{\tau} = 2\vec{r} \times \vec{F} \rightarrow \tau = -2rbIB \sin(\theta) = 0$$

$$B \text{ emf} = \frac{d\Phi}{dt} = -2rbB\omega \sin(\theta) = 0$$

The magnetic field generated by the loop interacts with the field from the permanent magnet: the fields are aligned such that the plane of the conducting loop is perpendicular to the field of the permanent magnet – no torque is developed.



Conducting loop in a magnetic field

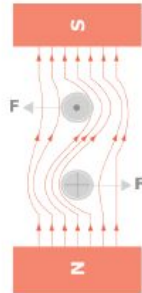


$$\theta = \omega t = \frac{\pi}{2}$$

$$\Phi = 0$$

$$\tau = -2rbIB = -\tau_{\text{MAX}}$$

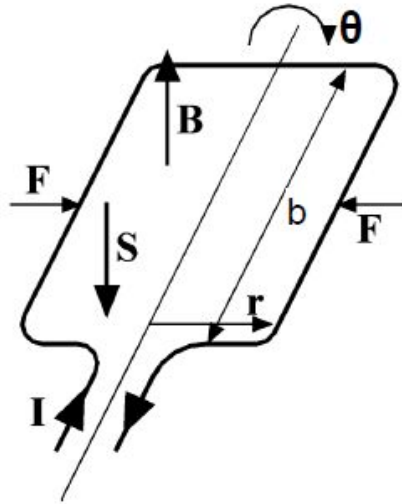
$$Bemf = -2rbB\omega = -Bemf_{\text{MAX}}$$



The magnetic field generated by the conducting loop interacts with the field from the permanent magnet in such a way as to generate a torque on the loop.



Conducting loop in a magnetic field



$$\theta = \omega t = \pi$$

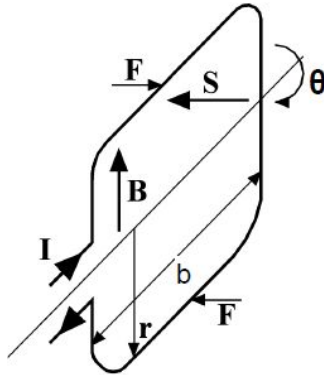
$$\Phi = -2rbBI = -\Phi_{\text{MAX}}$$

$$\tau = 0$$

$$B_{\text{emf}} = 0$$



Conducting loop in a magnetic field



$$\theta = \omega t = \frac{3}{2}\pi$$

$$\Phi = 0$$

$$\tau = 2rbIB = \tau_{\text{MAX}}$$

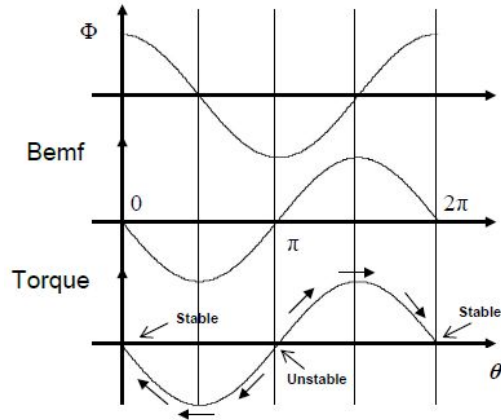
$$\text{Bemf} = 2rbLB\omega = \text{Bemf}_{\text{MAX}}$$

Note: $\tau_{\text{MAX}} \propto I$

$$\text{Bemf}_{\text{MAX}} \propto \omega$$



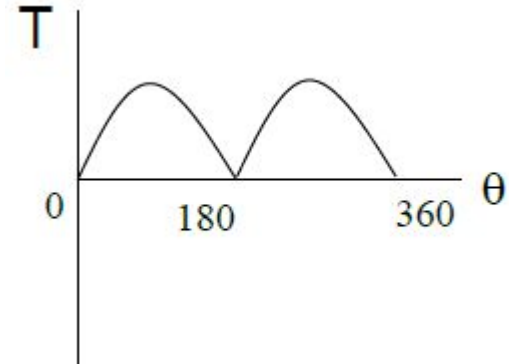
Conducting loop in a magnetic field



$$\Phi = \Phi_{\text{MAX}} \cos \theta$$

$$Bemf = -Bemf_{\text{MAX}} \sin \theta$$

$$\tau = \tau_{\text{MAX}} \sin \theta$$

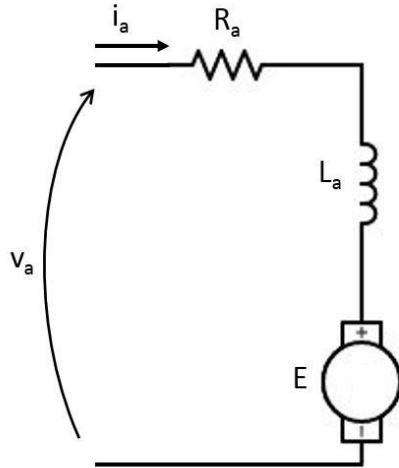


If current always flows in the same direction, then the loop will only oscillate rather than rotate. Here come in play the commutator-brushes system, which reverse the direction of the current on each half cycle in order to make the rotor rotate.

Now the torque is always applied in the same direction inducing the loop to spin continuously.



DC Machine – Equivalent circuit



The excitation circuit is implied, it is hidden in the electromotive force at the ends of the brushes (index of the rotation of the rotor inside the stator poles).

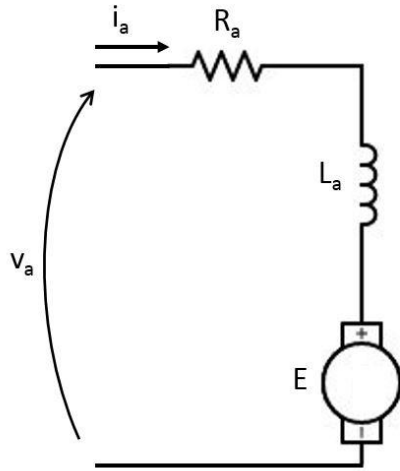
The electromotive force E represents an active load.

R_a is the resistance of the armature winding (L_a , armature inductance, in steady state does not come into play) and the resistance associated with the collector brush contact.

V_a is the armature voltage, I_a the armature current, K_e the machine constant, ω the angular velocity, Φ the excitation flux per pole.



DC Machine – Equivalent circuit simplified



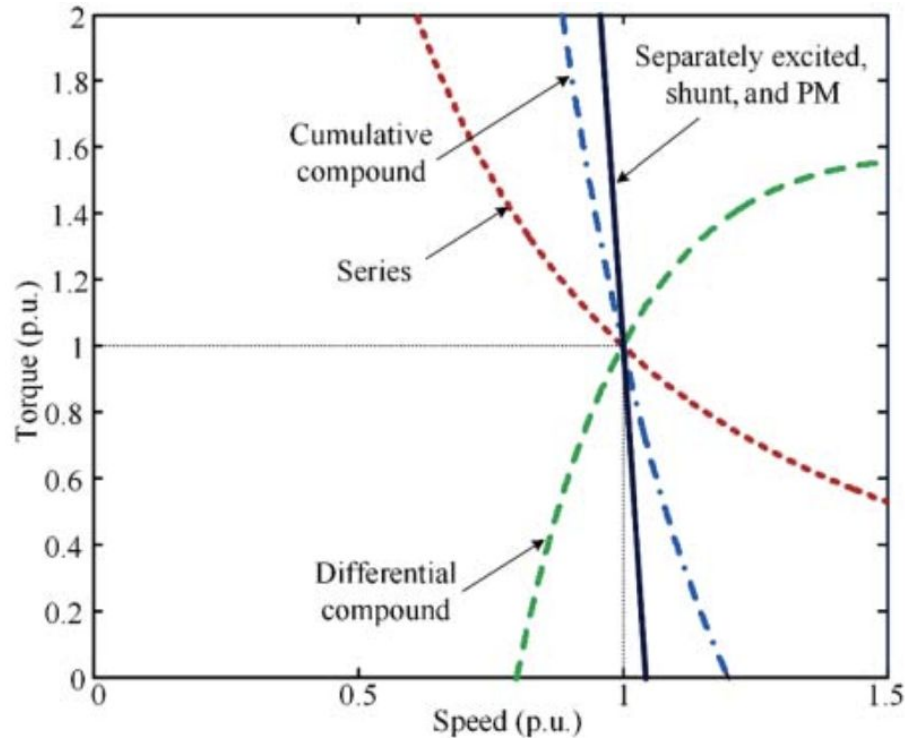
$$\begin{cases} V_a = R_a I_a + E \\ E = K_e \phi \omega \\ \tau = K_e \phi I_a \end{cases}$$

$$I_a = \frac{V_a - E}{R_a} = \frac{V_a - K_e \phi \omega}{R_a}$$

$$\tau = K_e \phi I_a = K_e \phi \frac{V_a - K_e \phi \omega}{R_a} = \frac{K_e \phi}{R_a} V_a - \frac{(K_e \phi)^2}{R_a} \omega$$



Torque – speed characteristics





Speed regulation

$$\omega = \frac{V_a - R_a I_a}{K_e \phi}$$

There are three ways to adjust the speed:

- Act on the armature voltage (separated excitation only)
- Act on the flow of excitation
- Act on the armature resistance

These actions can be carried out individually or jointly, each being effective in a different speed range. The armature current is controlled when you want to have the maximum output torque of the machine.



Speed regulation – Armature voltage

$$\tau = \frac{K_e \phi}{R_a} V_a - \frac{(K_e \phi)^2}{R_a} \omega$$

By varying the voltage the constant term varies, which means that the speed when the torque is zero depends on V. If V reduces the point of intersection of the characteristic with the axis **ω is reduced**, it moves, but the characteristic remains parallel (the characteristics translate parallel to the nominal one). If we increase the V, the **ω increases**.



Speed regulation – Flow of excitation

Reducing the flow results in an increase in speed.

If the flow is reduced, the E decreases and with $V = V_n$ the I_a increases.

In the torque expression $\tau = K_e \phi I_a$, ϕ decreases and I_a increases; I_a prevails, the torque increases and the motor accelerates.

$$I_a = \frac{V - E}{R_a}$$

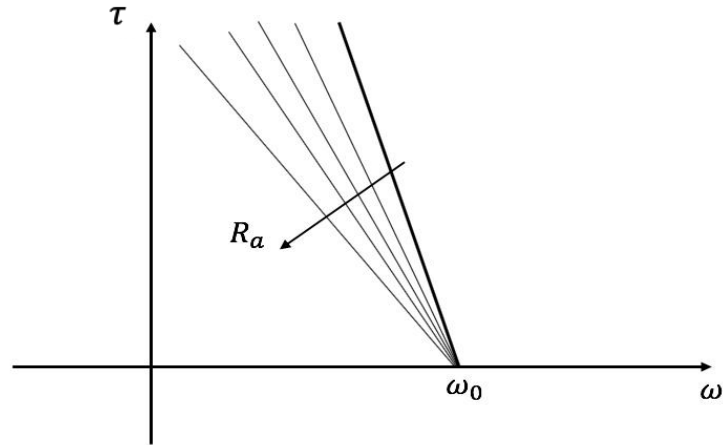
Since the R_a is very small, the increase in I_a is enormously greater than the reduction in flow. Thus, the torque increases and the machine accelerates.

Conversely, **if ϕ is increased**, E increases and I_a decreases, the torque decreases and the machine decelerates (brakes).



Speed regulation – Armature resistance

$$\tau = \frac{K}{R_a} V_a - \frac{K^2}{R_a} \omega \quad \omega = \frac{V}{K} - \frac{R_a}{K^2} \tau$$

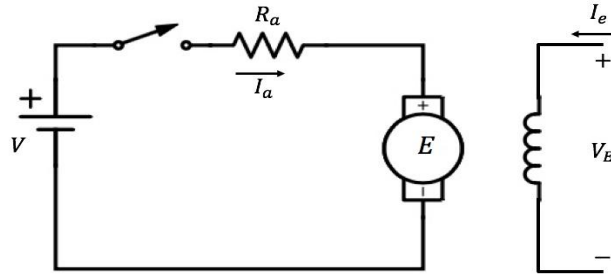




DC Generator



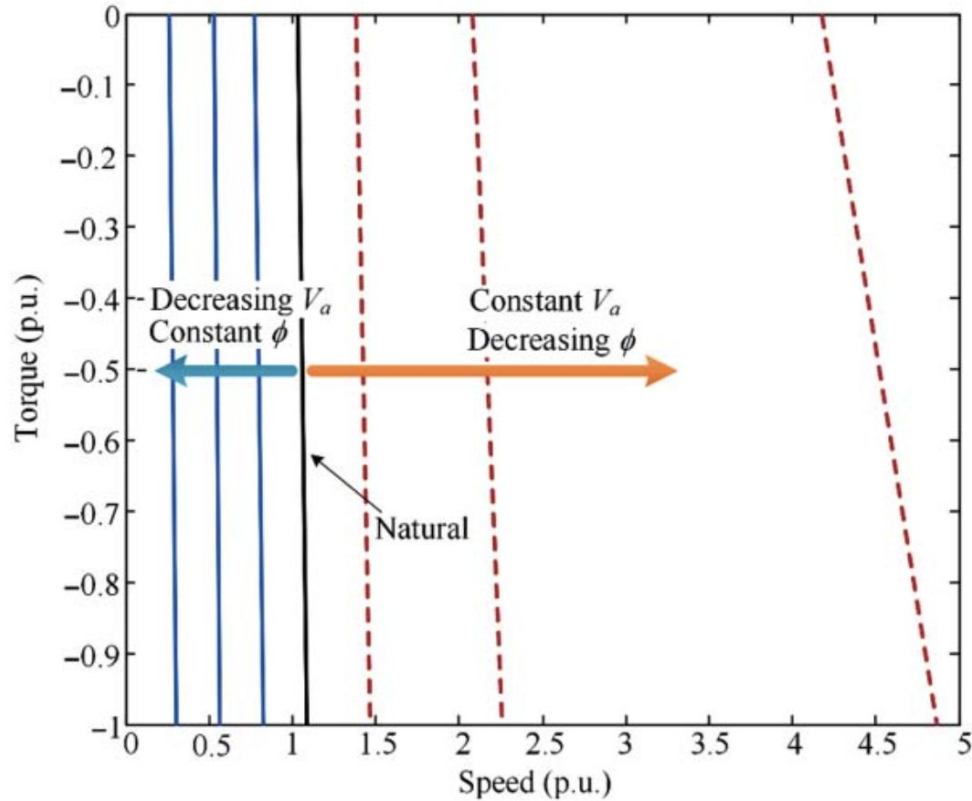
DC Machine - Breaking



If the directions of the voltages and currents are those shown in the figure, the machine acts as a motor (switch closed). If the E is made greater than V , the armature current is reversed, the excitation flux has remained with its polarity and the machine switches to operate as a **brake** because the torque reverses becoming negative and instead of being a driving force it becomes braking. Therefore, the machine acts as a brake towards the load and from the electrical point of view it acts as a **generator**, takes energy from the load, converts it into electricity and this can be dissipated or recovered.

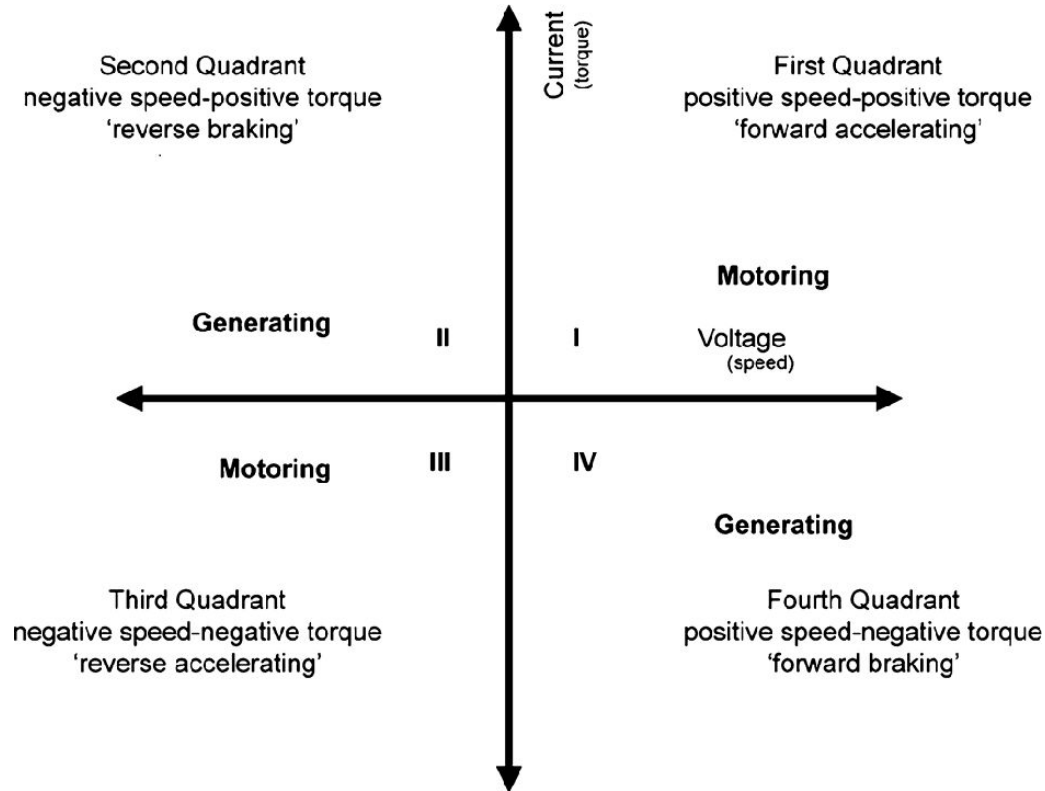


DC Machine - Breaking





DC Machine – 4 quadrant operation





Conclusions



DC Machine

Pros:

- Simple operation and motor control
- Cheapest and simplest motor
- Speed proportional to voltage

Cons:

- Requires high maintenance
- Sparking
- Low-life span (due to physical wear on brushes)



DC motor drive technology is very mature and well developed. However, with the ever-increasing demand on high efficiency, high power density, and maintenance-free operation, DC motor drives are **no longer attractive** for EV propulsion.

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